Note on vector potential expansion 3/7/2011

(1) Consider the following integral

$$T = \int \vec{J}(\vec{r}\,') \cdot \vec{\nabla}' F(\vec{r}\,')\, dV'$$

gradient with respect to variable $\vec{r}\,'$.

where $\vec{J}(\vec{r}\,')$ is current density (localized and time-independent)

$F(\vec{r}\,')$ = scalar function

Use product rule (5): $\vec{\nabla}\cdot(F\vec{J}) = F\,\vec{\nabla}\cdot\vec{J} + \vec{J}\cdot\vec{\nabla}F$

Hence $T = \int \vec{\nabla}'\cdot(F\vec{J})\, dV' - \int F(\vec{r}\,')\underbrace{\vec{\nabla}'\cdot\vec{J}(\vec{r}\,')}_{0}\, dV'$

For a stationary current continuity equation

$$\vec{\nabla}\cdot\vec{J} + \frac{\partial\rho}{\partial t} = 0 \quad \text{but } \frac{\partial\rho}{\partial t} = 0 \Rightarrow \vec{\nabla}\cdot\vec{J} = 0$$

By divergence theorem

$$T = \oint_{\text{area}} F\vec{J}\cdot d\vec{s} = 0$$

because we can always choose S such that $\vec{J}(\vec{r}\in S) = 0$, so that S encloses all localized current.

Hence $\boxed{T=0}$ for arbitrary function $F(\vec{r}\,')$

(2) Take now $F(\vec{r}\,') = \vec{a}\cdot\vec{r}\,'$ where $\vec{a}$ = constant vector

$$\vec{\nabla}'(\vec{a}\cdot\vec{r}\,') = \left(\hat{x}\frac{\partial}{\partial x'} + \hat{y}\frac{\partial}{\partial y'} + \hat{z}\frac{\partial}{\partial z'}\right)(a_x x' + a_y y' + a_z z')$$

$$= \hat{x}a_x + \hat{y}a_y + \hat{z}a_z = \vec{a}$$

Hence $T = \int \vec{J}(\vec{r}\,')\cdot\vec{a}\, dV' = \vec{a}\cdot\int\vec{J}(\vec{r}\,')\, dV' = 0$

for any vector $\vec{a}$ $\Rightarrow$ must be that $\int\vec{J}(\vec{r}\,')\, dV = 0$ (eq. 0)

(3) Now take $F(\vec{r}\,') = (\vec{a}\cdot\vec{r}\,')(\vec{r}\cdot\vec{r}\,')$

Use rule (#3)
$$\vec{\nabla}' F(\vec{r}\,') = (\vec{a}\cdot\vec{r}\,')\,\vec{\nabla}'(\vec{r}\cdot\vec{r}\,') + (\vec{r}\cdot\vec{r}\,')\,\vec{\nabla}'(\vec{a}\cdot\vec{r}\,')$$
$$= (\vec{a}\cdot\vec{r}\,')\,\vec{r} + (\vec{r}\cdot\vec{r}\,')\,\vec{a}$$

Hence. $T = \int \vec{J}(\vec{r}\,')\cdot\left[\vec{r}\,(\vec{a}\cdot\vec{r}\,') + \vec{a}\,(\vec{r}\cdot\vec{r}\,')\right]dV' = 0$

$$\Rightarrow \int \vec{J}(\vec{r}\,')\cdot\vec{r}\,(\vec{r}\,'\cdot\vec{a})\,dV' = -\int \vec{J}(\vec{r}\,')\cdot\vec{a}\,(\vec{r}\cdot\vec{r}\,')\,dV'$$

Hence (keep track of dots! $\Rightarrow$ they represent scalar product of two vectors), since $\vec{a}$ = constant,

$$\vec{a}\cdot\int \left(\vec{r}\cdot\vec{J}(\vec{r}\,')\right)\vec{r}\,'\,dV' = -\vec{a}\cdot\int \vec{J}(\vec{r}\,')(\vec{r}\cdot\vec{r}\,')\,dV'$$

Since $\vec{a}$ is arbitrary, the two vectors must be equal:

$$\text{(eq.1)}\quad \int\left(\vec{r}\cdot\vec{J}(\vec{r}\,')\right)\vec{r}\,'\,dV' = -\int \vec{J}(\vec{r}\,')(\vec{r}\cdot\vec{r}\,')\,dV'$$

(4) Now we consider combination like this:

$$\vec{r}\,'\times\vec{J}(\vec{r}\,')\times\vec{r} = \vec{J}(\vec{r}\,')(\vec{r}\,'\cdot\vec{r}) - \vec{r}\,(\vec{r}\,'\cdot\vec{J}(\vec{r}\,'))$$
$$\vec{A}\times\vec{B}\times\vec{C} = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$$

and $\vec{r}\times\vec{J}(\vec{r}\,')\times\vec{r}\,' = \vec{J}(\vec{r}\,')(\vec{r}\cdot\vec{r}\,') - \vec{r}\,'(\vec{r}\cdot\vec{J}(\vec{r}\,'))$

Thus, integrate both sides over volume V':

$$\int \vec{r}\times\vec{J}(\vec{r}\,')\times\vec{r}\,'\,dV' = \int \vec{J}(\vec{r}\,')(\vec{r}\cdot\vec{r}\,')\,dV' - \int \vec{r}\,'(\vec{r}\cdot\vec{J}(\vec{r}\,'))\,dV'$$

Using (eq.1) we see that second integral on the right-hand side equals first integral $\Rightarrow$

(eq.2) $$\frac{1}{2}\int \vec{r}\times\vec{J}(\vec{r}\,')\times\vec{r}\,'\,dV' = \int \vec{J}(\vec{r}\,')\,(\vec{r}\cdot\vec{r}\,')\,dV'$$

$$\Rightarrow \int \vec{J}(\vec{r}\,')\,(\vec{r}\cdot\vec{r}\,')\,dV' = \vec{r}\times\left(-\frac{1}{2}\int \vec{r}\,'\times\vec{J}(\vec{r}\,')\,dV'\right)$$

$$= -\vec{r}\times\vec{m} = \vec{m}\times\vec{r}$$

where $\vec{m} \equiv \frac{1}{2}\int \vec{r}\,'\times\vec{J}(\vec{r}\,')\,dV'$ (eq.3)

is magnetic moment of current distribution.

For the vector potential it means:

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\,dV' = \left(\text{expansion in powers } \frac{r'}{r}\right)$$

$$\cong \frac{\mu_0}{4\pi}\underbrace{\int \frac{\vec{J}(\vec{r}\,')}{r}\,dV'}_{0 \text{ by (eq.0)}} + \frac{\mu_0}{4\pi r^3}\underbrace{\int (\vec{r}\cdot\vec{r}\,')\,\vec{J}(\vec{r}\,')\,dV'}_{\text{use (eq.2)}}$$

$$= \frac{\mu_0}{4\pi r^3}\,\vec{m}\times\vec{r} = \frac{\mu_0}{4\pi r^2}\,\vec{m}\times\hat{r}$$

where $\vec{m}$ is given by (eq.3)